



2012

TRIAL

HIGHER SCHOOL CERTIFICATE
EXAMINATION
GIRRAWEEEN HIGH SCHOOL

MATHEMATICS EXTENSION 2

General Instructions

- Reading time -- 5 minutes
- Working time -- 3 hours
- Write using black or blue pen
- Board -- approved calculators may be used
- A table of standard integrals is provided
- Show all necessary working in

Questions 11-16

Total marks - 100

Section 1 pages 2-3

10 marks

- Attempt Questions 1-10
- Allow about 20 minutes for this section

Section 2 pages 4 - 11

- Attempt Questions 11 - 16
- Allow about 2 hours 40 minutes for this section

2

SECTION 1

Multiple Choice (10 marks) Circle your answer on the question paper.

- If z_1 and z_2 are any two non-zero complex numbers such that $|z_1 + z_2| = |z_1| + |z_2|$ then $\arg(z_1) - \arg(z_2)$ is
(A) $-\frac{\pi}{2}$ (B) 0 (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$
- If z is a complex number such that $|z - 3 - 4i| + |z + 3 + 4i| = 10$, then the locus of z is
(A) An ellipse (B) a circle (C) a hyperbola (D) a straight line
- The real values of x and y if $\sqrt{x}(i + \sqrt{y}) - 15 = i(8 - \sqrt{y})$
(A) 36, 225 (B) 25, 9 (C) 25, 225 (D) 9, 25
- If α and β are the roots of $x^2 - 2x + 4 = 0$, then the value of $\alpha^3 + \beta^3$ is
(A) $\frac{5}{2}$ (B) -128 (C) -16 (D) 64
- If $\int x^4 \sin(6x^5) dx = \frac{\lambda}{6} \cos(6x^5) + C$, $x \neq 0$ then the value of λ is
(A) 5 (B) $\frac{1}{5}$ (C) -5 (D) $-\frac{1}{5}$
- If $\phi(x) = \int_0^x t \sin t dt$, then $\phi'(x)$ is
(A) $x \cos x$ (B) $x \sin x$ (C) $\cos x + x \sin x$ (D) $\frac{x^2}{2}$

7. The value of $\int_0^2 |x-1| dx$ is

- (A) -1 (B) 1 (C) 2 (D) 3

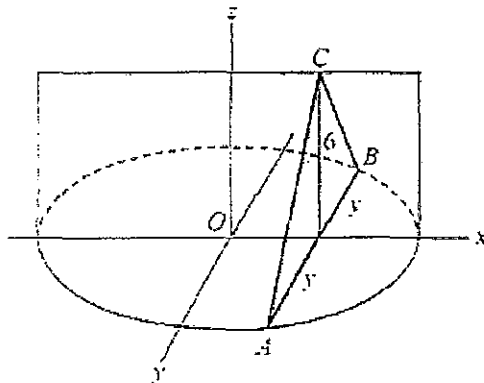
8. The coordinates of a focus of the ellipse $4x^2 + 9y^2 = 1$ are

- (A) $\left(-\frac{\sqrt{5}}{6}, 0\right)$ (B) $\left(0, \frac{\sqrt{5}}{6}\right)$ (C) $\left(\frac{\sqrt{5}}{3}, 0\right)$ (D) $\left(-\frac{\sqrt{5}}{3}, 0\right)$

9. The equations of the directrices of the hyperbola $3x^2 - 6y^2 = -18$ are

- (A) $x = \pm 1$ (B) $y = \pm 1$ (C) $x = \pm 2$ (D) $y = \pm 2$

10. A solid has a base in the form of an ellipse with major axis 10 and minor axis 8. Every cross-section perpendicular to the major axis is an isosceles triangle with altitude 6. Which one of the following is the correct expression for the volume of the solid.



- (A) $V = \frac{24}{5} \int_{-4}^4 \sqrt{25-y^2} dy$ (C) $V = \frac{24}{5} \int_{-5}^5 \sqrt{25-x^2} dx$
 (B) $V = 4 \int_0^4 \sqrt{25-y^2} dy$ (D) $V = 4 \int_0^5 \sqrt{25-x^2} dx$

Question 11 (15 marks)

Marks

Evaluate:

(a) $\int \frac{e^x - e^{-x}}{(e^x + e^{-x})^2} dx$ 2

(b) $\int x \tan^{-1} x dx$ 2

(c) $\int_0^{\frac{\pi}{2}} \frac{dx}{2 + \cos x}$ 3

(d) (i) Find the real numbers A, B and C such that

$$\frac{3x+1}{(x-2)^2(x+2)} = \frac{A}{x-2} + \frac{B}{(x-2)^2} + \frac{C}{x+2}$$
 2

(ii) Hence evaluate $\int \frac{3x+1}{(x-2)^2(x+2)} dx$ 2

(e) Use the result $\int_0^a f(x) dx = \int_0^a f(a-x) dx$ to evaluate $\int_0^{\frac{\pi}{4}} \log(1 + \tan x) dx$ 4

Question 12 (15 marks)

(a) (i) Prove that if $x = \alpha$ is a root of multiplicity k of the real polynomial equation

$$P(x) = 0, \text{ then } x = \alpha \text{ is also a root of the equation } \frac{dP}{dx} = 0 \text{ of multiplicity } k-1. \quad 2$$

(ii) Solve $P(x) = x^4 - 11x^3 + 42x^2 - 68x + 40$, given that $P(x) = 0$ has a root of multiplicity 3. 2

(b) Let α, β, γ be the roots of the cubic equation $x^3 + px^2 + q = 0$, where p, q are real.

The equation $x^3 + ax^2 + bx + c = 0$ has roots $\alpha^2, \beta^2, \gamma^2$. Find a, b, c in terms of p, q .

2

- (c) A particle of mass m kilograms starts falling from rest having been initially projected vertically upwards from the ground. It experiences air resistance of magnitude mkv^2 on both the upward and downward motion of the journey where k is a positive constant and v is the velocity of the particle at any instant.

(i) Show that the terminal velocity, V_0 , of the particle is given by $V_0 = \sqrt{\frac{g}{k}}$. 1

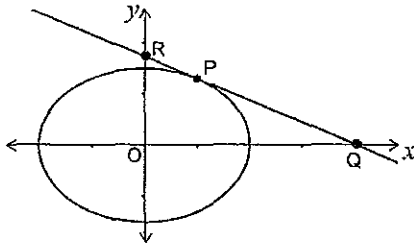
- (ii) If W is the velocity of the particle when it hits the ground, show that the distance,

S , fallen is given by $\frac{1}{2k} \ln \left(\frac{g}{g - kW^2} \right)$. 2

(iii) The maximum height attained by the particle is given by $H = \frac{1}{2k} \ln \left(\frac{g + kU^2}{g} \right)$

where U is the initial velocity of projection. Show that $\frac{1}{W^2} = \frac{1}{U^2} + \frac{1}{V_0^2}$. 2

- (d) The point $P(4\cos\theta, 3\sin\theta)$ lies on the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$.



- (i) Find the equation of the tangent to the ellipse at P. 2

- (ii) The tangent at P cuts the x -axis at Q and the y -axis at R. Show that the area of

ΔORQ is $\frac{12}{\sin 2\theta}$. 1

- (iii) Find the coordinates of P so that area of ΔORQ is a minimum. 1

Question 13 (15 marks)

- (a) z is a complex number such that $|z| = 4$, $\arg z = \frac{5\pi}{6}$. Express z in the form

$a + ib$ where a and b are real. 1

- (b) $z_1 = 1 + i\sqrt{3}$ and $z_2 = 1 - i$ are two complex numbers.

- (i) Express z_1, z_2 and $z_1 z_2$ in modulus/argument form. 2

- (ii) Find the smallest positive integer such that $z_1^n z_2^n$ is purely imaginary. For this value of n , write the value of $z_1^n z_2^n$ in the form bi where b is a real number. 3

- (c) Sketch the locus of the following:

(i) $\arg(z - 1 - 2i) = \frac{\pi}{4}$ 1

(ii) $z\bar{z} - 3(z + \bar{z}) \leq 0$ 2

- (d) (i) Find the seven seventh roots of -1 2

(ii) Factorise $z^7 + 1$ over the real field R . 2

(iii) Prove that $\cos \frac{\pi}{7} + \cos \frac{3\pi}{7} + \cos \frac{5\pi}{7} = \frac{1}{2}$ 2

Question 14 (15 marks)

- (a) Find the volume of the solid generated by revolving the region bounded by $x^2 - y^2 = 16$, and $x = 8$ about the y -axis. (see Figure 1)

3

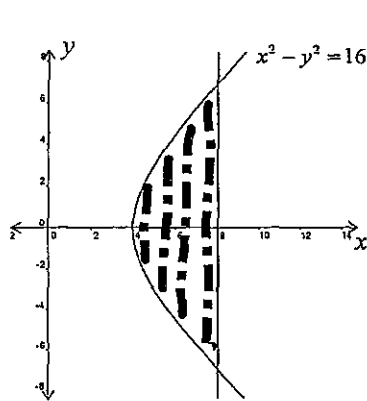


Figure 1

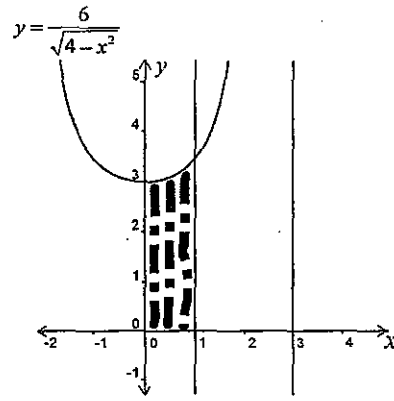


Figure 2

- (b) A mould for a section of concrete piping is made by rotating the region bounded by the curve $y = \frac{6}{\sqrt{4-x^2}}$ and the x -axis between the lines $x = 0$ and $x = 1$ through one complete revolution about the line $x = 3$. All measurements are in metres. (see Figure 2)

- (i) By considering strips of width Δx parallel to the axis of rotation, show that the volume $V \text{ m}^3$ of the concrete used in the piping is given by

$$V = 12\pi \int_0^1 \frac{3-x}{\sqrt{4-x^2}} dx \quad 2$$

- (ii) Hence find the volume of the concrete used in the piping, giving your answer

correct to the nearest cubic metre.

3

- (c) $P\left(p, \frac{1}{p}\right)$ and $Q\left(q, \frac{1}{q}\right)$ are two points on the rectangular hyperbola $xy = 1$. M is the midpoint of the chord PQ .

- (i) Show that the chord PQ has equation $x + pqy - (p + q) = 0$ 2

- (ii) If P and Q move on the rectangular hyperbola such that the perpendicular distance of the chord PQ from the origin $O(0,0)$ is always $\sqrt{2}$, show that

$$(p + q)^2 = 2(1 + p^2 q^2) \quad 1$$

- (iii) Hence find the equation of the locus of M , stating any restriction on its domain and range. 4

Question 15 (15 marks)

- (a) A body is projected vertically upwards from the surface of the Earth with initial speed u . The acceleration due to gravity at any point on its path is inversely proportional to the square of its distance from the centre of the Earth.

- (i) Prove that the speed at any position x is given by

$$v^2 = u^2 + 2gR^2 \left(\frac{1}{x} - \frac{1}{R} \right) \quad 2$$

- (ii) Prove that the greatest height H above the Earth's surface is given by

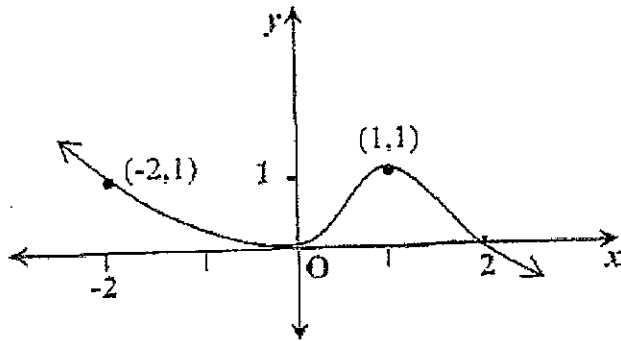
$$H = \frac{u^2 R}{2gR - u^2} \quad 2$$

- (iii) Show that the body will escape from the Earth if $u \geq \sqrt{2gR}$ 1

- (iv) If $u = \sqrt{2gR}$, prove that the time taken to reach a height $15R$ above the

$$\text{surface of the Earth is } 42\sqrt{\frac{R}{2g}} \quad 2$$

(b) The diagram shows the graph of $y = f(x)$. On separate diagrams sketch the graphs of the following. Clearly indicate any asymptotes and intercepts with the axes.



(i) $y = \ln[f(x)]$

2

(ii) $y = \frac{1}{f(x)}$

2

(iii) $y = -|f(x)|$

2

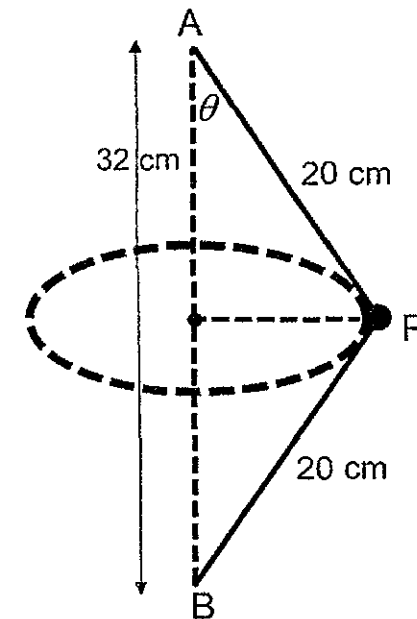
(c) Sketch the curve showing vertical and slant asymptotes.

$$f(x) = \frac{x^2 - 3x - 4}{x + 3}$$

2

Question 16 (15 marks)

(a) A particle P of mass m kg is tied to the midpoint of a light inextensible string of length 40 cm. One end of the string is fixed at point A , and the other end is fixed at point B which is 32 cm vertically below A . Particle P moves with constant speed v m/s in a horizontal circle around the midpoint of AB , while both sections of string AP and BP remain taut. The acceleration due to gravity is g m/s².



(i) Draw a diagram showing the forces acting on the particle P .

2

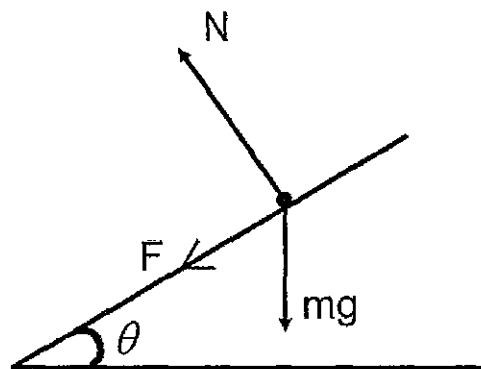
(ii) Find the tension in each part of the string in terms of m , v and g .

4

(iii) Show that $v \geq \frac{3}{10}\sqrt{g}$, for both strings to be taut.

1

- (b) A car travels at a uniform speed of $v \text{ m/s}$ around a banked circular track. The track is inclined at an angle θ to the horizontal and the car moves in a horizontal circle of radius r . The car experiences a normal reaction force, N , from the track, a vertical force of magnitude mg due to gravity and a sideways frictional force, F , acting down the slope. This information is shown in the diagram below.



- (i) Resolve the forces along the slope and perpendicular to the slope, or otherwise. Hence, find expressions for F and N . 4
- (ii) A track of radius 200 metres is banked at angle of 25° to the horizontal. Find the speed of cars around this track if there is no sideways friction force. Assume that $g = 9.8 \text{ m/s}$. 2
- (iii) A motorist is riding around the track at 90 km/h . Find the frictional force experienced by the motorist and in what direction. The combined mass of the car and motorist is 1500 kg . 2

END OF TEST

Trial HSC Extensions 2, 2012 - Solutions

Multiple choice (10 marks)

1 B 2 A 3 D 4 C 5 D 6 B 7 B

8 A 9 B 10 C

Question 11 (15 marks)

(a) $\int \frac{e^x - e^{-x}}{(e^x + e^{-x})^2} dx$

Let $u = e^x + e^{-x}$

$\frac{du}{dx} = e^x - e^{-x}$

$du = (e^x - e^{-x}) dx$

$I = \int \frac{du}{u^2} = \int u^{-2} du$

$= \frac{u^{-1}}{-1} + C$

$= -\frac{1}{u} + C$

$= -\frac{1}{e^x + e^{-x}} + C$

(b) $\int x \tan^{-1} x dx$

$= \tan^{-1} x \cdot \frac{x^2}{2} - \int \frac{1}{1+x^2} \times \frac{x^2}{2} dx$

$= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int \frac{x^2}{1+x^2} dx \quad \text{--- (1)}$

$\int \frac{x^2}{x^2+1} dx = \int \left(\frac{x^2+1-1}{x^2+1} \right) dx$

$= \int \frac{x^2+1}{x^2+1} dx - \int \frac{1}{x^2+1} dx$

$= x - \tan^{-1} x + C$

① becomes

$I = \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} (x - \tan^{-1} x + C)$

$= \frac{x^2}{2} \tan^{-1} x - \frac{x}{2} + \frac{1}{2} \tan^{-1} x + C$

(c) $\int_0^{\frac{\pi}{2}} \frac{dx}{2+\cos x}$

Let $t = \tan \frac{x}{2}$ and $\cos x = \frac{1-t^2}{1+t^2}$

$\frac{dx}{dt} = \left(\sec^2 \frac{x}{2} \right) \frac{1}{2}$

$= \frac{1}{2} (1 + \tan^2 \frac{x}{2})$

$dx = \frac{1}{2} (1+t^2) dt$

$dx = \frac{2 dt}{1+t^2}$

When $x=0$, $t=0$

When $x = \frac{\pi}{2}$, $t = \tan \frac{\pi}{4} = 1$

$2 + \cos x = 2 + \frac{1-t^2}{1+t^2}$

$= \frac{2(1+t^2) + 1-t^2}{1+t^2}$

$= \frac{2+2t^2+1-t^2}{1+t^2}$

$= \frac{3+t^2}{1+t^2}$

$I = \int_0^1 \frac{1+t^2}{3+t^2} \times \frac{2 dt}{1+t^2}$

$= \int_0^1 \frac{2 dt}{3+t^2}$

$= 2 \int_0^1 \frac{dt}{t^2 + (\sqrt{3})^2}$

$2 \left[\frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{t}{\sqrt{3}} \right) \right]_0^1$

$= \frac{2}{\sqrt{3}} \left[\tan^{-1} \left(\frac{1}{\sqrt{3}} \right) \right]_0^1$

$= \frac{2}{\sqrt{3}} \left(\tan^{-1} \left(\frac{1}{\sqrt{3}} \right) - \tan^{-1}(0) \right)$

$= \frac{2}{\sqrt{3}} \left(\frac{\pi}{6} - 0 \right)$

$= \frac{2\pi}{6\sqrt{3}} = \frac{\pi}{3\sqrt{3}}$

(d) let $3x+1 \equiv A(x-2)(x+2) + B(x+2) + C(x-2)^2$

$x=2 \Rightarrow 7=4B \quad B=\frac{7}{4}$

$x=-2 \Rightarrow -5=16C \quad C=\frac{-5}{16}$

Comparing coefficients of x^2 on both sides of the

identity $A+C=0 \quad A=\frac{5}{16}$

$$\frac{3x+1}{(x-2)^2(x+2)} = \frac{5}{16} \times \frac{1}{x-2} + \frac{7}{4} \times \frac{1}{(x-2)^2} - \frac{5}{16(x+2)}$$

$$\int \frac{(3x+1)dx}{(x-2)^2(x+2)} = \frac{5}{16} \int \frac{dx}{x-2} + \frac{7}{4} \int \frac{dx}{(x-2)^2} - \frac{5}{16} \int \frac{dx}{x+2}$$

$$= \frac{5}{16} \log(x-2) - \frac{7}{4(x-2)} - \frac{5}{16} \log(x+2) + C$$

(e) $\frac{\pi}{4} \int_0^{\frac{\pi}{4}} \log(1+\tan x) dx = \frac{\pi}{4} \int_0^{\frac{\pi}{4}} \log\left(\frac{1+\tan x + 1-\tan x}{1+\tan x}\right) dx$

$$= \frac{\pi}{4} \int_0^{\frac{\pi}{4}} \log\left(1 + \tan\left(\frac{\pi}{4} - x\right)\right) dx = \frac{\pi}{4} \int_0^{\frac{\pi}{4}} \log \frac{2}{1+\tan x} dx$$

$$= \frac{\pi}{4} \int_0^{\frac{\pi}{4}} \log\left(1 + \frac{\tan \frac{\pi}{4} - \tan x}{1 + \tan \frac{\pi}{4} \cdot \tan x}\right) dx = \frac{\pi}{4} \int_0^{\frac{\pi}{4}} (\log 2 - \log(1+\tan x)) dx$$

$$= \frac{\pi}{4} \int_0^{\frac{\pi}{4}} \log 2 dx - \frac{\pi}{4} \int_0^{\frac{\pi}{4}} \log(1+\tan x) dx$$

$$\frac{\pi}{4} \int_0^{\frac{\pi}{4}} \log(1+\tan x) dx + \frac{\pi}{4} \int_0^{\frac{\pi}{4}} \log(1+\tan x) dx = \frac{\pi}{4} \int_0^{\frac{\pi}{4}} \log 2 dx$$

$$\frac{\pi}{2} \int_0^{\frac{\pi}{4}} \log(1+\tan x) dx = \log 2 \left[x \right]_0^{\frac{\pi}{4}} \\ = \log 2 \left(\frac{\pi}{4} - 0 \right) \\ = \frac{\pi}{4} \log 2$$

$$\frac{\pi}{4} \int_0^{\frac{\pi}{4}} \log(1+\tan x) dx = \frac{\pi \log 2}{4}$$

Question 12 (15 marks)

(a)(i) let $p(x) = (x-a)^k \cdot Q(x)$ where $Q(a) \neq 0$

$$\begin{aligned} \frac{dp}{dx} &= k(x-a)^{k-1} Q(x) + (x-a)^k \frac{dQ}{dx} \\ &= (x-a)^{k-1} \left[k Q(x) + (x-a) \frac{dQ}{dx} \right] \\ &= (x-a)^{k-1} S(x) \end{aligned}$$

where $S(x) = k Q(x) + (x-a) \frac{dQ}{dx}$ is a polynomial

and $S(a) \neq 0$

$\therefore x=a$ is a root of multiplicity $k-1$ of the equation $\frac{dp}{dx} = 0$

(ii) Since $p(x)$ has a root of multiplicity 3, $p(x), p'(x)$ and $p''(x)$ have a common zero.

$$p'(x) = 4x^3 - 11 \times 3x^2 + 42x - 68$$

$$= 4x^3 - 33x^2 + 84x - 68$$

$$p''(x) = 4 \times 3x^2 - 33 \times 2x + 84$$

$$= 12x^2 - 66x + 84$$

$$p''(x) = 0 \Rightarrow 12x^2 - 66x + 84 = 0$$

$$6(2x^2 - 11x + 14) = 0$$

$$2x^2 - 11x + 14 = 0$$

$$\textcircled{-7, -4} \quad pq = 28$$

$$p + q = -11$$

$$2x^2 - 4x - 7x + 14 = 0$$

$$2x(x-2) - 7(x-2) = 0$$

$$(x-2)(2x-7) = 0$$

$$x = 2 \quad \text{or} \quad x = \frac{7}{2}$$

$$p'(2) = 4 \times 8 - 33 \times 4 + 84 \times 2 - 68$$

$$= 200 - 200 = 0$$

$$p(2) = 16 - 11 \times 8 + 42 \times 4 - 68 \times 2 + 40$$

$$= 16 - 88 + 168 - 136 + 40$$

$$= 0$$

Sum of the roots

$$x_1, 2, 2, 2$$

$$6 + x = 11$$

$$x = 5$$

Roots are 2, 2, 2, 5

$$(b) x^3 + px^2 + q = 0$$

$$y = x^2 \quad x = \sqrt{y}$$

$$(\sqrt{y})^3 + p(\sqrt{y})^2 + q = 0$$

$$y^{\frac{3}{2}} + py + q = 0$$

$$y^{\frac{3}{2}} = -(py + q)$$

$$y^3 = (py + q)^2$$

$$y^3 = p^2 y^2 + 2pqy + q^2$$

$$y^3 - p^2 y^2 - 2pqy - q^2 = 0$$

$$x^3 - p^2 x^2 - 2pqx - q^2 = 0$$

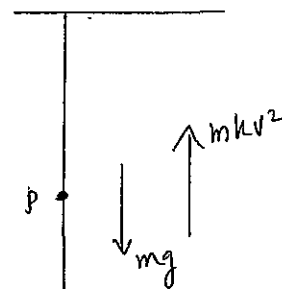
$$x^3 + ax^2 + bx + c = 0$$

$$a = -p^2$$

$$b = -2pq$$

$$c = -q^2$$

(c)



Equation of motion is

$$m\ddot{x} = mg - mkv^2$$

$$\ddot{x} = g - kv^2$$

Terminal velocity V_0 occurs when $\ddot{x} = 0$

$$g - kV_0^2 = 0$$

$$kV_0^2 = g; V_0^2 = \frac{g}{k}$$

$$V_0 = \sqrt{\frac{g}{k}}$$

$$(ii) v \frac{dv}{dx} = g - kv^2$$

$$\frac{dw}{dx} = \frac{g - kv^2}{v}$$

$$\frac{dx}{dv} = \frac{v}{g - kv^2}$$

$$dx = \frac{v dv}{g - kv^2}$$

$$[x]_0^S = \int_0^w \frac{v dv}{g - kv^2}$$

$$S = \int_0^w \frac{-2k v dv}{-2k(g - kv^2)}$$

$$= \frac{-1}{2k} \left[\log(g - kv^2) \right]_0^w$$

$$= \frac{-1}{2k} (\log(g - kw^2) - \log g)$$

$$= \frac{1}{2k} \log\left(\frac{g}{g - kw^2}\right)$$

(iii) $S = H$

$$\frac{1}{2k} \log \left(\frac{g}{g - kw^2} \right) = \frac{1}{2k} \log \left(\frac{g + ku^2}{g} \right)$$

$$\frac{g}{g - kw^2} = \frac{g + ku^2}{g}$$

$$(g - kw^2)(g + ku^2) = g^2$$

$$g^2 + gku^2 - gkw^2 - k^2u^2w^2 = g^2$$

$$gku^2 - gkw^2 - k^2u^2w^2 = 0$$

divide by U^2W^2gk

$$\frac{gku^2}{U^2W^2gk} - \frac{gkw^2}{U^2W^2gk} - \frac{k^2u^2w^2}{U^2W^2gk} = 0$$

$$\frac{1}{W^2} - \frac{1}{U^2} - \frac{k}{g} = 0$$

$$\frac{1}{W^2} - \frac{1}{U^2} = \frac{k}{g}$$

$$\underline{\underline{\frac{1}{W^2} = \frac{1}{U^2} + \frac{1}{V_0^2}}}$$

(d) $x = 4\cos\theta$, $y = 3\sin\theta$

$$\frac{dx}{d\theta} = 4 \times -\sin\theta = -4\sin\theta$$

$$\frac{dy}{d\theta} = 3\cos\theta$$

$$\frac{dy}{dx} = \frac{dy}{d\theta} \times \frac{d\theta}{dx}$$

$$= \frac{3\cos\theta}{-4\sin\theta}$$

Equation of tangent

$$y - 3\sin\theta = \frac{-3\cos\theta}{4\sin\theta} (x - 4\cos\theta)$$

$$4\sin\theta y - 12\sin^2\theta = -3x\cos\theta + 12\cos^2\theta$$

$$4y\sin\theta + 3x\cos\theta = 12$$

$$\frac{4y\sin\theta}{12} + \frac{3x\cos\theta}{12} = 1$$

$$\underline{\underline{\frac{x\cos\theta}{4} + \frac{y\sin\theta}{3} = 1}}$$

(ii) $x=0 \Rightarrow \frac{y\sin\theta}{3} = 1$

$$y = \frac{3}{\sin\theta}$$

$$y=0 \Rightarrow \frac{x\cos\theta}{4} = 1$$

$$x = \frac{4}{\cos\theta}$$

Area of ΔORQ

$$= \frac{1}{2} \times OQ \times OR$$

$$= \frac{1}{2} \times \frac{4}{\cos\theta} \times \frac{3}{\sin\theta} = \frac{6}{\sin\theta\cos\theta}$$

$$= \frac{12}{2\sin\theta\cos\theta}$$

$$= \underline{\underline{\frac{12}{\sin 2\theta}}}$$

(iii) $\frac{12}{\sin 2\theta}$ is minimum when $\sin 2\theta$ is maximum

$$\text{i.e. } \sin 2\theta = 1$$

$$2\theta = \frac{\pi}{2} \quad \theta = \frac{\pi}{4}$$

The coordinates of P are

$$\left(4\cos\frac{\pi}{4}, 3\sin\frac{\pi}{4} \right)$$

$$= \left(4 \times \frac{1}{\sqrt{2}}, 3 \times \frac{1}{\sqrt{2}} \right)$$

$$= \underline{\underline{\left(\frac{4}{\sqrt{2}}, \frac{3}{\sqrt{2}} \right)}}$$

Question 13 (15 marks)

$$(a) z = 4 \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right)$$

$$= 4 \left(-\frac{\sqrt{3}}{2} + \frac{1}{2}i \right)$$

$$= \underline{\underline{-2\sqrt{3} + 2i}}$$

$$(b)(i) z_1 = 1 + i\sqrt{3}$$

$$r = \sqrt{1^2 + (\sqrt{3})^2} = 2$$

$$\tan \alpha = \frac{\sqrt{3}}{1} = \sqrt{3}$$

$$\alpha = \frac{\pi}{3}$$

$$\text{argument } \theta = \alpha = \frac{\pi}{3}$$

$$1 + i\sqrt{3} = 2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$$

$$z_2 = 1 - i$$

$$r = \sqrt{1^2 + (-1)^2} = \sqrt{2}$$

$$\tan \alpha = 1$$

$$\alpha = \frac{\pi}{4}$$

$$\theta = -\alpha = -\frac{\pi}{4}$$

$$1 - i = \sqrt{2} \left(\cos \frac{-\pi}{4} + i \sin \frac{-\pi}{4} \right)$$

$$|z_1 z_2| = |z_1| |z_2| = 2\sqrt{2}$$

$$\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$$

$$= \frac{\pi}{3} - \frac{\pi}{4} = \frac{\pi}{12}$$

$$z_1^n z_2^n = (z_1 z_2)^n$$

$$= (2\sqrt{2})^n \left[\cos \frac{n\pi}{12} + i \sin \frac{n\pi}{12} \right]$$

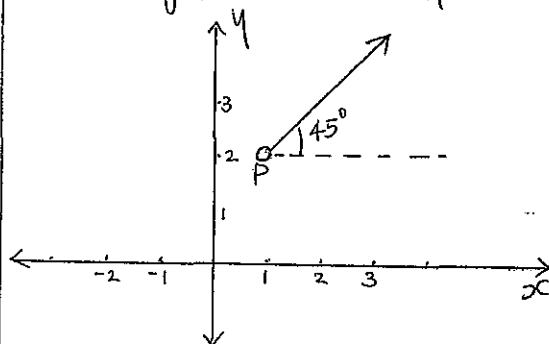
$z_1^n z_2^n$ is purely imaginary
means $\arg(z_1^n z_2^n)$ is a multiple
of $\frac{\pi}{2}$

$$\text{when } n=6, \arg(z_1^n z_2^n) = \frac{\pi}{2}$$

$$z_1^6 z_2^6 = (2\sqrt{2})^6 \left[\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right]$$

$$= \underline{\underline{512i}}$$

$$(c)(i) \arg(z - (1 + 2i)) = \frac{\pi}{4}$$



The locus is the half ray at
P(1, 2), making 45° to the
x axis, the point P is not
included.

$$(ii) z\bar{z} - 3(z + \bar{z}) \leq 0$$

$$\text{If } z = x + iy \text{ then } \bar{z} = x - iy$$

$$z\bar{z} = (x + iy)(x - iy)$$

$$= x^2 + y^2$$

$$z + \bar{z} = x + iy + x - iy$$

$$= 2x$$

$$z\bar{z} - 3(z + \bar{z})$$

$$= x^2 + y^2 - 6x$$

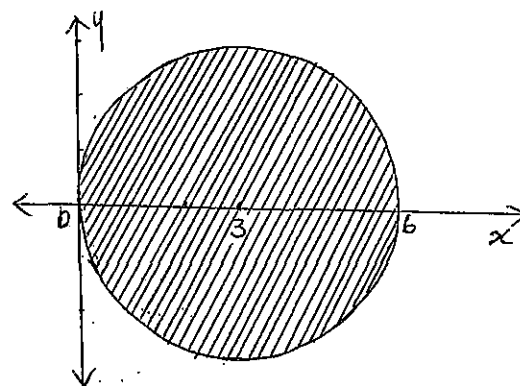
$$= x^2 - 6x + y^2$$

$$= x^2 - 6x + 9 - 9 + y^2$$

$$(x - 3)^2 + y^2 - 9 \leq 0$$

$$(x - 3)^2 + y^2 \leq 9$$

The locus is all points
that are on and inside
the circle of radius 3 units
and centre at (3, 0)



$$(d)(i) z^7 = -1 = \cos \pi + i \sin \pi$$

$$= \cos(2k\pi + \pi) + i \sin(2k\pi + \pi)$$

$$k = 0, 1, 2, \dots$$

The seven seventh roots of -1

are given by

$$z = \left[\cos(2k\pi + \pi) + i \sin(2k\pi + \pi) \right]$$

$$= \cos \frac{(2k+1)\pi}{7} + i \sin \frac{(2k+1)\pi}{7}$$

$$\text{where } k = 0, 1, 2, 3, 4, 5, 6$$

by De Moivre's theorem.

$$k=0$$

$$z_1 = \cos \frac{\pi}{7} + i \sin \frac{\pi}{7}$$

$$k=1$$

$$z_2 = \cos \frac{3\pi}{7} + i \sin \frac{3\pi}{7}$$

$$k=2$$

$$z_3 = \cos \frac{5\pi}{7} + i \sin \frac{5\pi}{7}$$

$$k=3$$

$$z_4 = \cos \frac{7\pi}{7} + i \sin \frac{7\pi}{7} = -1$$

$$k=4$$

$$z_5 = \cos \frac{9\pi}{7} + i \sin \frac{9\pi}{7}$$

$$k=5$$

$$z_6 = \cos \frac{11\pi}{7} + i \sin \frac{11\pi}{7}$$

$$k=6$$

$$z_7 = \cos \frac{13\pi}{7} + i \sin \frac{13\pi}{7}$$

$$(ii) Z_7 = \overline{Z_1}$$

$$Z_1 + Z_7 = 2 \cos \frac{\pi}{7}$$

$$Z_6 = \overline{Z_2}$$

$$Z_2 + Z_6 = 2 \cos \frac{3\pi}{7}$$

$$Z_5 = \overline{Z_3}$$

$$Z_3 + Z_5 = 2 \cos \frac{5\pi}{7}$$

$$Z^7 + 1 = (Z - Z_1)(Z - Z_2)(Z - Z_3)(Z - Z_4)(Z - Z_5)(Z - Z_6)(Z - Z_7)$$

$$= (Z+1)(Z - Z_1)(Z - Z_7)(Z - Z_2)(Z - Z_6)(Z - Z_3)(Z - Z_5)$$

$$= (Z+1)(Z^2 - Z(Z_1 + Z_7) + 1)(Z^2 - Z(Z_2 + Z_6) + 1)(Z^2 - Z(Z_3 + Z_5) + 1)$$

$$= (Z+1)(Z^2 - Z \times 2 \cos \frac{\pi}{7} + 1)(Z^2 - Z \times 2 \cos \frac{3\pi}{7} + 1)(Z^2 - Z \times 2 \cos \frac{5\pi}{7} + 1) \quad \text{--- (1)}$$

Also

$$Z^7 + 1 = (Z+1)(Z^6 - Z^5 + Z^4 - Z^3 + Z^2 - Z + 1) \quad \text{--- (2)}$$

From (1) and (2) we get-

$$(Z^2 - 2 \cos \frac{\pi}{7} Z + 1)(Z^2 - 2 \cos \frac{3\pi}{7} Z + 1)(Z^2 - 2 \cos \frac{5\pi}{7} Z + 1)$$

$$= Z^6 - Z^5 + Z^4 - Z^3 + Z^2 - Z + 1$$

$$Z^4 - 2 \cos \frac{3\pi}{7} Z^3 + Z^2 - 2 \cos \frac{\pi}{7} Z^3 + 4 \cos \frac{\pi}{7} \cos \frac{3\pi}{7} Z^2$$

$$- 2 \cos \frac{\pi}{7} Z + Z^2 - 2 \cos \frac{3\pi}{7} Z + 1)(Z^2 - 2 \cos \frac{5\pi}{7} Z + 1)$$

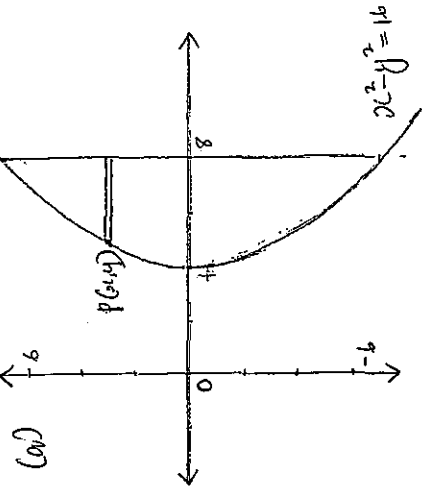
$$= Z^6 - Z^5 + Z^4 - Z^3 + Z^2 - Z + 1$$

Equating coefficients of Z^5 on both sides

$$- 2 \cos \frac{5\pi}{7} Z^5 - 2 \cos \frac{3\pi}{7} Z^5 - 2 \cos \frac{\pi}{7} Z^5 = -Z^5$$

$$2 \cos \frac{\pi}{7} + 2 \cos \frac{3\pi}{7} + 2 \cos \frac{5\pi}{7} = 1$$

Question 14 (15 marks)



Substitute $x = 8$ in

$$x^2 - y^2 = 16$$

$$64 - y^2 = 16$$

$$y^2 = 48$$

$$y = \pm \sqrt{48}$$

$$= \pm 4\sqrt{3}$$

Area of annulus

$$= \pi(8^2 - x^2)$$

Volume of annulus

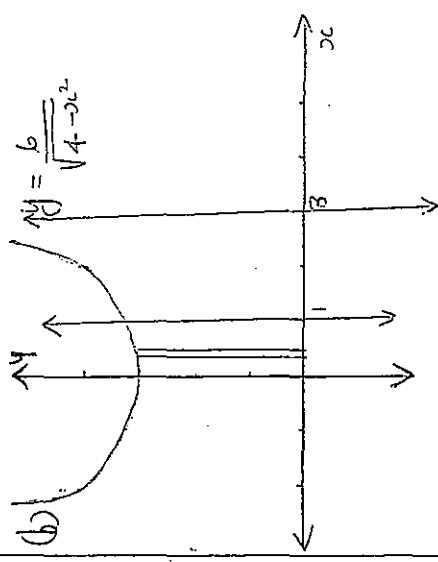
$$\Delta V = \pi(8^2 - x^2) \Delta y$$

Volume of solid

$$V = \lim_{\Delta y \rightarrow 0} \sum_{4\sqrt{3}}^{4\sqrt{3}} \pi(64 - x^2) \Delta y$$

$$= \int_{-4\sqrt{3}}^{4\sqrt{3}} \pi(64 - x^2) dy$$

$$\begin{aligned} &= \int_{-4\sqrt{3}}^{4\sqrt{3}} \pi(64 - 16 - y^2) dy \\ &= 2\pi \int_{-4\sqrt{3}}^{4\sqrt{3}} (48 - y^2) dy \\ &= 2\pi \left[48y - \frac{y^3}{3} \right]_{-4\sqrt{3}}^{4\sqrt{3}} \\ &= 2\pi \left\{ (48 \times 4\sqrt{3} - \frac{64 \times 3\sqrt{3}}{3}) - \left(-48 \times 4\sqrt{3} + \frac{64 \times 3\sqrt{3}}{3} \right) \right\} \\ &= 2\pi (192\sqrt{3} - 64\sqrt{3}) \\ &= 2\pi \times 128\sqrt{3} \\ &= 256\sqrt{3} \pi \text{ cubic units.} \end{aligned}$$



Radius = $3 - x$

height = y

Volume of cylindrical shell

$$\Delta V = 2\pi r h \Delta x$$

$$= 2\pi(3 - x)y \Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^1 \frac{2\pi(3-x) \times 6}{\sqrt{4-x^2}} \Delta x$$

$$= \int_0^1 \frac{2\pi(3-x) \times 6}{\sqrt{4-x^2}} dx$$

$$= 12\pi \int_0^1 \frac{3-x}{\sqrt{4-x^2}} dx$$

$$(ii) V = 12\pi \int_0^1 \frac{3}{\sqrt{4-x^2}} dx - 12\pi \int_0^1 \frac{x}{\sqrt{4-x^2}} dx$$

$$= 36\pi \int_0^1 \frac{dx}{\sqrt{4-x^2}} - 12\pi \int_0^1 \frac{x}{\sqrt{4-x^2}} dx$$

$I_1 \qquad I_2$

$$I_1 = 36\pi \left[\sin^{-1} \frac{x}{2} \right]_0^1$$

$$= 36\pi (\sin^{-1} \frac{1}{2} - \sin^{-1} 0)$$

$$= 36\pi \times \frac{\pi}{6} = 6\pi^2$$

$$I_2 = \int_0^1 \frac{x}{\sqrt{4-x^2}} dx$$

Let $u = 4-x^2$

$$\frac{du}{dx} = -2x$$

$$-2x dx = du$$

$$x dx = -\frac{du}{2}$$

when $x=0$, $u=4$

when $x=1$, $u=3$

$$= \int_4^3 \frac{-\frac{du}{2}}{\sqrt{u}} = -\frac{1}{2} \int_4^3 \frac{du}{\sqrt{u}}$$

$$= \frac{1}{2} \int_3^4 u^{-\frac{1}{2}} du$$

$$= \frac{1}{2} \left[\frac{u^{\frac{1}{2}}}{\frac{1}{2}} \right]_3^4 = \frac{1}{2} \times 2 \left[u^{\frac{1}{2}} \right]_3^4$$

$$= 4^{\frac{1}{2}} - 3^{\frac{1}{2}} = 2 - \sqrt{3}$$

$$I_2 = 12\pi (2 - \sqrt{3})$$

$$= 24\pi - 12\sqrt{3}\pi$$

$$V = 6\pi^2 - 24\pi + 12\sqrt{3}\pi$$

$$= \underline{\underline{49\pi^3}}$$

$$(c) P(p, \frac{1}{p}) \quad Q(q, \frac{1}{q})$$

$$m_{PQ} = \frac{\frac{1}{p} - \frac{1}{q}}{p - q}$$

$$= \frac{q-p}{pq} \times \frac{1}{p-q} = -\frac{1}{pq}$$

Equation of PQ

$$y - \frac{1}{p} = -\frac{1}{pq} (x - p)$$

$$pqy - q = -x + p$$

$$x + pqy - q - p = 0$$

$$\underline{\underline{x + pqy - (p+q) = 0}}$$

$$(ii) \frac{|0 + 0 - (p+q)|}{\sqrt{1^2 + (pq)^2}} = \sqrt{2}$$

$$\frac{|p+q|}{\sqrt{1+p^2q^2}} = \sqrt{2}$$

$$\frac{(p+q)^2}{1+p^2q^2} = 2$$

$$\underline{\underline{(p+q)^2 = 2(1+p^2q^2)}}$$

$$(iii) M = \left(\frac{p+q}{2}, \frac{\frac{1}{p} + \frac{1}{q}}{2} \right)$$

$$= \left(\frac{p+q}{2}, \frac{p+q}{2pq} \right)$$

$$x = \frac{p+q}{2} \quad y = \frac{p+q}{2pq}$$

$$x^2 = \frac{(p+q)^2}{4} \quad y^2 = \frac{(p+q)^2}{4p^2q^2}$$

$$\frac{x^2}{y^2} = \frac{(p+q)^2}{4} \times \frac{4p^2q^2}{(p+q)^2}$$

$$\frac{x^2}{y^2} = p^2q^2 \quad \text{--- (1)}$$

From (ii) $(p+q)^2 = 2(1+p^2q^2)$

$$\frac{(p+q)^2}{2} = 1 + p^2q^2$$

$$\frac{(2x)^2}{2} = 1 + p^2q^2$$

$$\frac{4x^2}{2} = 1 + p^2q^2$$

$$2x^2 - 1 = p^2q^2$$

(1) becomes $\frac{x^2}{y^2} = 2x^2 - 1$

$$x^2 = 2x^2y^2 - y^2 = y^2(2x^2 - 1)$$

$$\underline{\underline{y^2 = \frac{x^2}{2x^2 - 1}}}$$

which is the locus of M.

Domain of $y^2 = \frac{x^2}{2x^2-1}$

$$2x^2 - 1 > 0$$

$$2x^2 > 1$$

$$x^2 > \frac{1}{2}$$

$$x > \frac{1}{\sqrt{2}} \text{ or } x < -\frac{1}{\sqrt{2}}$$

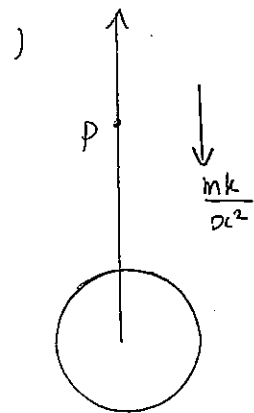
$$D: \left\{ x : |x| > \frac{1}{\sqrt{2}} \right\}$$

$y^2 = \frac{x^2}{2x^2-1}$ is symmetric

in x and y . Hence the

range $\{y : |y| > \frac{1}{\sqrt{2}}\}$

Question 15 (15 marks)



Equation of motion is

$$m\ddot{x} = -\frac{mk}{x^2} \quad \text{--- (1)}$$

Substitute $\ddot{x} = v \frac{dv}{dx}$

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(1) becomes

$$m v \frac{dv}{dx} = -\frac{mk}{x^2}$$

$$v \frac{dv}{dx} = -\frac{k}{x^2} \quad \text{--- (2)}$$

At the Earth's surface $x=R$ and

$$g = \frac{k}{R^2} \quad \therefore k = gR^2$$

(2) becomes

$$v \frac{dv}{dx} = -\frac{gR^2}{x^2}$$

$$v dv = -\frac{gR^2}{x^2} dx$$

$$\int v dv = -gR^2 \int \frac{dx}{x^2}$$

$$\frac{v^2}{2} = -gR^2 \times \frac{x^{-1}}{-1} + C$$

$$\frac{v^2}{2} = \frac{gR^2}{x} + C \quad \text{--- (3)}$$

when $x=R$, $v=u$

$$\frac{u^2}{2} = gR + C$$

$$C = \frac{u^2}{2} - gR$$

(3) becomes

$$\frac{v^2}{2} = \frac{gR^2}{x} + \frac{u^2}{2} - gR$$

$$v^2 = \frac{2gR^2}{x} + u^2 - \frac{2gR^2}{R}$$

$$v^2 = u^2 + 2gR^2 \left(\frac{1}{x} - \frac{1}{R} \right)$$

(ii) At the greatest height $v=0$

$$0 = u^2 + 2gR^2 \left(\frac{1}{x} - \frac{1}{R} \right)$$

$$2gR^2 \left(\frac{1}{x} - \frac{1}{R} \right) = -u^2$$

$$\frac{1}{x} - \frac{1}{R} = \frac{-u^2}{2gR^2}$$

$$\frac{1}{x} = \frac{1}{R} - \frac{u^2}{2gR^2}$$

$$\frac{1}{x} = \frac{2gR - u^2}{2gR^2}$$

$$x = \frac{2gR^2}{2gR - u^2}$$

The greatest height H

above the Earth is

$$H = \frac{2gR^2}{2gR - u^2} - R$$

$$= \frac{2gR^2 - R(2gR - u^2)}{2gR - u^2}$$

$$= \frac{2gR^2 - 2gR^2 + u^2R}{2gR - u^2}$$

$$= \frac{u^2R}{2gR - u^2}$$

(iii) If the particle escapes from the Earth, there is no maximum height, since the particle never turns downward again.

This is equivalent to saying $H \rightarrow \infty$

$$H \rightarrow \infty \Rightarrow 2gR - u^2 = 0$$

$$2gR = u^2$$

$$u^2 = \sqrt{2gR} \quad (\text{since } u > 0)$$

The body will escape the Earth if $u \geq \sqrt{2gR}$

(iv) $g = 10 \text{ m/s}^2$ $R = 6400 \text{ km}$

$$= \frac{10}{1000} \text{ km/s}^2$$

$$u = \sqrt{2 \times \frac{1}{1000} \times 6400} \text{ km/s}$$

$$= \sqrt{12.8} \text{ km/s}$$

$$= \underline{\underline{11.31 \text{ km/s}}}$$

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 (V) Let t_1 be the time taken by the body to rise to a height of $16R$ above the Earth's surface. During this time x changes from R to $16R$

$$V^2 = u^2 + 2gR^2 \left(\frac{1}{x} - \frac{1}{R} \right)$$

Substitute $u^2 = 2gR$

$$V^2 = 2gR + \frac{2gR^2}{x} - 2gR$$

$$V = \frac{\sqrt{2gR}}{x^{\frac{1}{2}}}$$

$$\frac{dx}{dt} = \frac{\sqrt{2gR}}{x^{\frac{1}{2}}}$$

$$\frac{dt}{dx} = \frac{x^{\frac{1}{2}}}{\sqrt{2gR}}$$

$$dt = \frac{x^{\frac{1}{2}}}{\sqrt{2gR}} dx$$

$$\int_0^{t_1} dt = \int_R^{16R} \frac{x^{\frac{1}{2}}}{\sqrt{2gR}} dx$$

$$= \frac{1}{\sqrt{2gR}} \left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}} \right]_R^{16R}$$

$$= \frac{1}{\sqrt{2gR}} \times \frac{2}{3} \left[x^{\frac{3}{2}} \right]_R^{16R}$$

$$= \frac{1}{\sqrt{2gR}} \times \frac{2}{3} \left((16R)^{\frac{3}{2}} - R^{\frac{3}{2}} \right)$$

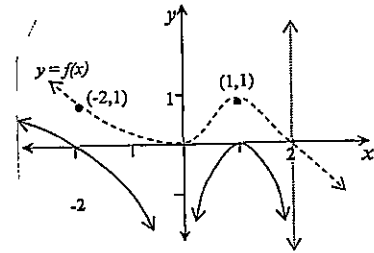
$$= \frac{1}{\sqrt{2gR}} \times \frac{2}{3} \left(64R^{\frac{3}{2}} - R^{\frac{3}{2}} \right)$$

$$= \frac{1}{\sqrt{2gR}} \times \frac{2}{3} \times 63R^{\frac{3}{2}}$$

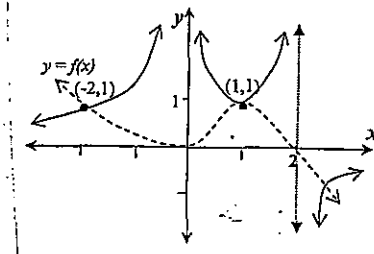
$$= \frac{1}{\sqrt{2g}} \times \frac{2}{3} \times 63R^{\frac{1}{2}}$$

$$= 42 \sqrt{\frac{R}{2g}}$$

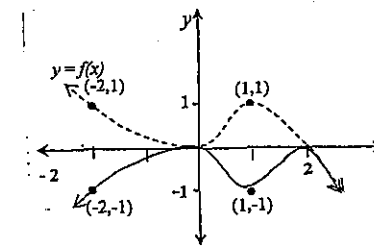
(b) (i) $y = \ln[f(x)]$



(ii) $y = \frac{1}{f(x)}$



(iii) $y = -|f(x)|$



(c) $f(x) = \frac{x^2 - 3x - 4}{x + 3}$

Vertical asymptote: $x = -3$

$$f(x) = 0 \Rightarrow x^2 - 3x - 4 = 0$$

$$(x - 4)(x + 1) = 0$$

$$x = 4 \text{ or } -1$$

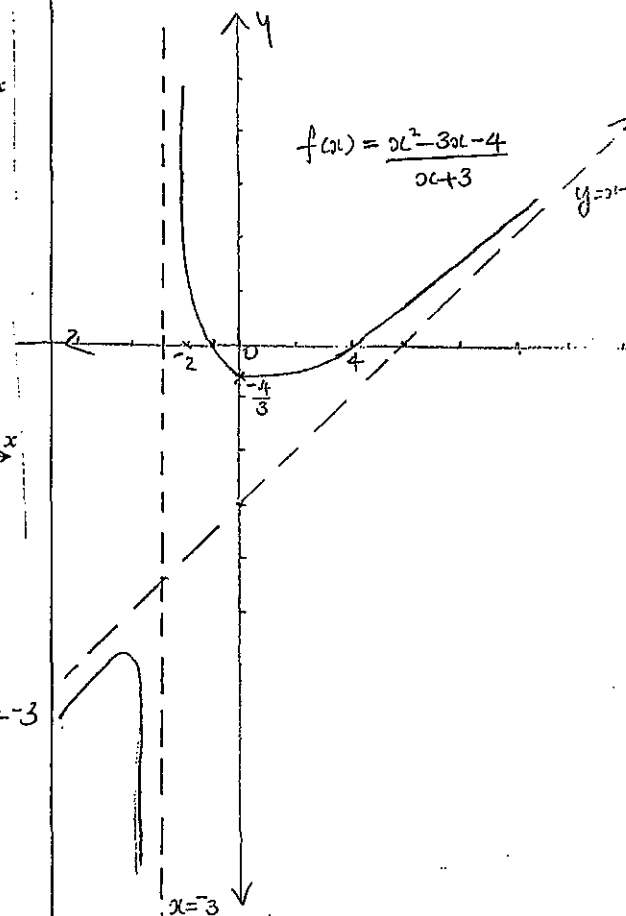
x intercepts are $(4, 0)$ $(-1, 0)$

$$x = 0 \Rightarrow f(x) = \frac{-4}{3}$$

y intercept $(0, -\frac{4}{3})$

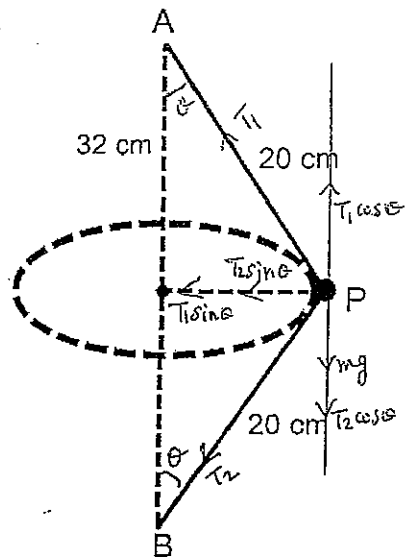
$$\begin{array}{r} x-6 \\ x+3 \overline{) x^2-3x-4} \\ \underline{x^2+3x} \\ -6x-4 \\ \underline{-6x-18} \\ 14 \end{array}$$

Slant asymptote is $y = x - 6$



Question 16 (15 marks)

a) (i)



ii) Resolving the forces at P

Vertically

$$T_1 \cos \theta = T_2 \cos \theta + mg$$

$$T_1 \cos \theta - T_2 \cos \theta = mg$$

$$(T_1 - T_2) \cos \theta = mg$$

$$T_1 - T_2 = \frac{mg}{\cos \theta}$$

$$\cos \theta = \frac{16}{20} = \frac{4}{5}$$

$$T_1 - T_2 = \frac{5}{4} mg$$

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Horizontally

$$T_1 \sin \theta + T_2 \sin \theta = \frac{mv^2}{r}$$

$$(T_1 + T_2) \sin \theta = \frac{mv^2}{0.12}$$

$$T_1 + T_2 = \frac{mv^2}{0.12} \times \frac{1}{\sin \theta}$$

$$\sin \theta = \frac{12}{20} = \frac{3}{5}$$

$$T_1 + T_2 = \frac{mv^2}{0.12} \times \frac{5}{3} = \frac{5}{0.36} mv^2$$

$$T_1 - T_2 = \frac{5}{4} mg \quad \text{--- (1)}$$

$$T_1 + T_2 = \frac{5}{0.36} mv^2 \quad \text{--- (2)}$$

adding (1) and (2)

$$2T_1 = \frac{5}{4} mg + \frac{5}{0.36} mv^2$$

$$= m \left(\frac{5}{4} g + \frac{5}{0.36} v^2 \right)$$

(2) - (1) gives

$$2T_2 = \frac{5}{0.36} mv^2 - \frac{5}{4} mg$$

$$= m \left(\frac{5v^2}{0.36} - \frac{5}{4} g \right)$$

$$T_1 = \frac{m}{2} \left(\frac{5}{4} g + \frac{5}{0.36} v^2 \right)$$

$$T_2 = \frac{m}{2} \left(\frac{5v^2}{0.36} - \frac{5}{4} g \right)$$

$$T_2 \geq 0$$

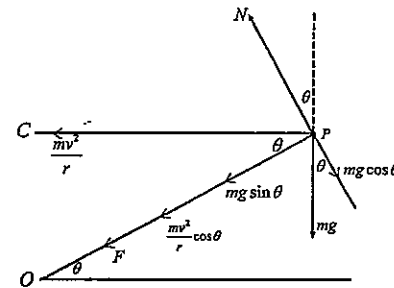
$$\frac{5v^2}{0.36} - \frac{5}{4} g \geq 0$$

$$v^2 \geq \frac{g}{4} \times 0.36$$

$$v \geq 0.3 \sqrt{g}$$

$$v \geq \frac{3}{10} \sqrt{g}$$

(b) (i)



Resolving along the slope

$$\frac{mv^2}{r} \cos \theta = F + mg \sin \theta$$

$$F = \frac{mv^2}{r} \cos \theta - mg \sin \theta$$

Resolving perpendicular to the slope

$$\frac{mv^2}{r} \sin \theta = N - mg \cos \theta$$

$$N = \frac{mv^2}{r} \sin \theta + mg \cos \theta$$

(ii) $F = 0$

$$\therefore \frac{mv^2}{r} \cos \theta = mg \sin \theta$$

$$\frac{v^2}{r} \cos \theta = g \sin \theta$$

$$\frac{v^2}{rg} = \tan \theta ; v^2 = rg \tan \theta$$

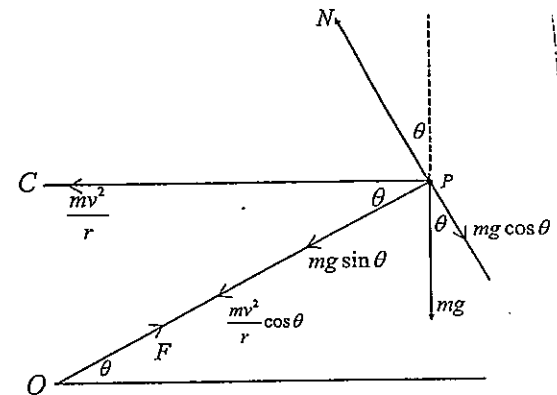
$$v = \sqrt{rg \tan \theta}$$

$$= \sqrt{200 \times 9.8 \times \tan 25^\circ}$$

$$= 30 \text{ m/s} = 108 \text{ km/h}$$

(iii) $90 \text{ km/h} = 25 \text{ m/s}$

The car is travelling at a speed less than the design speed of the track. The car tends to slip down the track. So friction will react to this by pushing up the slope.

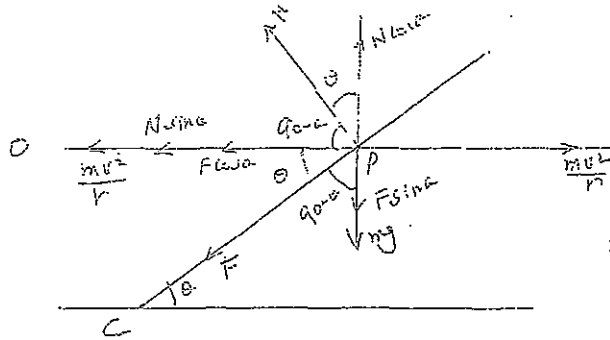


$$F = mg \sin \theta - \frac{mv^2}{r} \cos \theta$$

$$= 1500 \times 9.8 \times \sin 25^\circ - \frac{1500 \times 625}{200} \cos 25^\circ$$

$$= 1964 \text{ Newtons up the track}$$

Alternative solution - Resolving forces along horizontal and vertical direction.



horizontally:

$$N \sin \theta + F \cos \theta = \frac{mv^2}{r} \quad \text{--- (1)}$$

Vertically:

$$N \cos \theta - F \sin \theta = mg \quad \text{--- (2)}$$

$$\textcircled{1} \times \sin \theta \quad N \sin^2 \theta + F \sin \theta \cos \theta = \frac{mv^2}{r} \sin \theta \quad \text{--- (3)}$$

$$\textcircled{2} \times \cos \theta \quad N \cos^2 \theta - F \sin \theta \cos \theta = mg \cos \theta \quad \text{--- (4)}$$

$$\textcircled{3} + \textcircled{4} \quad N (\sin^2 \theta + \cos^2 \theta) = \frac{mv^2}{r} \sin \theta + mg \cos \theta$$

$$N = \frac{mv^2}{r} \sin \theta + mg \cos \theta$$

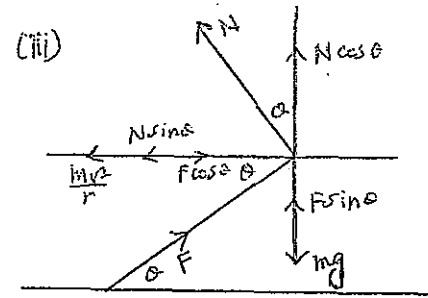
$$\textcircled{1} \times \cos \theta \quad N \sin \theta \cos \theta + F \cos^2 \theta = \frac{mv^2}{r} \cos \theta \quad \text{--- (5)}$$

$$\textcircled{2} \times \sin \theta \quad N \sin \theta \cos \theta - F \sin^2 \theta = mg \sin \theta \quad \text{--- (6)}$$

$$\textcircled{5} - \textcircled{6} \quad F \cos^2 \theta + F \sin^2 \theta = \frac{mv^2}{r} \cos \theta - mg \sin \theta$$

$$F = \frac{mv^2}{r} \cos \theta - mg \sin \theta$$

(ii) Same as the previous method



horizontally:

$$\frac{mv^2}{r} = N \sin \theta - F \cos \theta \quad \text{--- (1)}$$

Vertically:

$$mg = F \sin \theta + N \cos \theta \quad \text{--- (2)}$$

$$\textcircled{1} \times \cos \theta \quad N \sin \theta \cos \theta - F \cos^2 \theta = \frac{mv^2}{r} \cos \theta \quad \text{--- (3)}$$

$$\textcircled{2} \times \sin \theta \quad F \sin^2 \theta + N \sin \theta \cos \theta = mg \sin \theta \quad \text{--- (4)}$$

$$\textcircled{4} - \textcircled{3} \quad F \sin^2 \theta + F \cos^2 \theta = mg \sin \theta - \frac{mv^2}{r} \cos \theta$$

$$F = mg \sin \theta - \frac{mv^2}{r} \cos \theta$$

$$= 1500 \times 9.8 \times \sin 25^\circ - \frac{1500 \times 625}{200} \cos 25^\circ$$

$$= \underline{\underline{1964 \text{ Newtons up the track.}}}$$

